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Methods, circuits and systems for obtaining impedance or dielectric measurements of a material under test

Abstract

Certain disclosed methods include: transmitting an excitation signal into the MUT and transmitting a reference signal to a set of magnitude and phase (M/P) detectors; receiving the response signal; separately comparing a magnitude and phase for each of the excitation signal and the reference signal with corresponding detection ranges for a first one of the M/P detectors; separately comparing a magnitude and phase for each of the response signal and the reference signal with corresponding detection ranges for a second one of the M/P detectors; iteratively adjusting the excitation signal until the response signal has both a magnitude and a phase within the corresponding detection ranges for the second M/P detector; and iteratively adjusting the reference signal until the reference signal has both a magnitude and a phase within the corresponding detection ranges for the first and the second M/P detectors.

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Background/Summary

PRIORITY CLAIM (1) This application claims priority to U.S. patent application Ser. No. 18/110,014, filed Feb. 15, 2023, which itself claims priority to U.S. patent application Ser. No. 17/739,255, filed on May 9, 2022, which itself claims priority to U.S. patent application Ser. No. 17/229,958, filed on Apr. 14, 2021, which itself claims priority to U.S. Provisional Patent Application No. 63/010,791, filed on Apr. 16, 2020, the entire contents of each of which is herein incorporated by reference.

TECHNICAL FIELD

(1) The disclosure relates generally to circuits, systems, and methods for determining characteristics of a material under test (MUT) by generating and measuring electric signals at a specific frequency or over a range of frequencies to measure the dielectric properties of that MUT. The dielectric characteristics can then be correlated to a physical property or properties of the MUT such as density or moisture.

BACKGROUND

(2) The use of electrical impedance measurements to quantify physical characteristics of construction, manufacturing, and biological materials is the basis of an increasing number of techniques, including impedance tomography and impedance spectroscopy. An important factor in successfully characterizing an MUT is the ability to obtain accurate and repeatable measurements of electromagnetic properties of an MUT (e.g., the electrical impedance, admittance, capacitance, permittivity, etc.) of that MUT. These measured values are subsequently converted to information about the dielectric properties of the MUT. However, conventional approaches for obtaining electrical data for characterizing MUTs can be both inaccurate and insufficiently repeatable.

SUMMARY

(3) All examples and features mentioned below can be combined in any technically possible way.

(4) The present application presents an electronic circuit and measurement system for generating electric excitation signals at a specific frequency or over a range of frequencies that enable accurate measurements of response signals after excitation of the MUT. The transmitted and response signals can be used to compute the impedance and dielectric properties of the MUT.

(5) Particular approaches involve generating and measuring electric signals at a specific frequency or over a range of frequencies to measure the impedance or the dielectric signature of that MUT. The measurement of electrical impedance to quantify physical properties of construction, manufacturing, or biological materials is the basis of a variety of measurement techniques including impedance tomography and impedance spectroscopy. One requirement to successful characterization of an MUT is the accurate and repeatable measurements of the electrical impedance (or admittance) signature, which subsequently is converted to information about the dielectric properties of the MUT, which in turn can be correlated with physical (non-electrical) properties of the MUT. Other electromagnetic characteristics can be used to successfully characterize an MUT in various implementations. Various particular implementations include an electronic circuit, measurement system, and method for generating electric excitation signals at a specific frequency or over a range of frequencies that provide accurate measurements of the electric signals in response to this excitation of the MUT for computation of the impedance and dielectric signal properties of the MUT.

(6) The subject matter of U.S. Pat. Nos. 5,900,736, 6,414,497, 6,803,771, 7,219,024, 9,465,061, and 9,805,146; as well as: US Patent Publication No. 2009/0270756, US Patent Publication No. 2012/0130212; US Patent Publication No. 2013/0307564, US Patent Publication No. 2014/0266268, US Patent Publication No. 2014/0278300, US Patent Publication No. 2015/0137831, US Patent Publication No. 2015/0212026, US Patent Publication No. 2018/0128934, US Patent Publication No. 2018/0172612; Provisional U.S. Patent Application No. 62/161,9275 (filed on Jan. 25, 2018); and Provisional U.S. Patent Application No. 62/661,682 (filed on Apr. 24, 2018) describe impedance-related techniques for determining characteristics of materials, and are each incorporated by reference herein in its entirety.

(7) The methods, electronic circuits and systems of the present subject matter relate to the measurement of the impedance as it varies with the dielectric properties of the MUT, as well as electronic devices and/or components for performing such measurements at a specific frequency or over a range of frequencies, with provisions for the characterization of the excitation (also referred to as “transmit”), the response (also referred to as “received”), and reference signals to produce a

measured signal within a desired range of the electronic measuring components over the frequency range, based upon the magnitude or strength and phase shift of the measured signal for the specific frequency or range of frequencies.

(8) In some cases, a method of characterizing a response signal for detecting physical characteristics of a material under test (MUT) includes: transmitting an excitation signal into the MUT using a transmitting electrode on a sensor array and transmitting a reference signal to a set of magnitude and phase (M/P) detectors; receiving a response signal from the MUT via a receiving electrode on the sensor array based on the excitation signal; separately comparing a magnitude and phase for each of the excitation signal and the reference signal with corresponding detection ranges for a first one of the M/P detectors; separately comparing a magnitude and phase for each of the response signal and the reference signal with corresponding detection ranges for a second one of the M/P detectors; and iteratively adjusting the excitation signal until the response signal has both a magnitude and a phase within the corresponding detection ranges for the second M/P detector; and iteratively adjusting the reference signal until the reference signal has both a magnitude and a phase within the corresponding detection ranges for the first and the second M/P detectors.

(9) In additional cases, a system is configured to characterize a response signal for detecting physical characteristics of a material under test (MUT). In these cases, the system can include: a sensor array for communicating with the MUT; a set of magnitude and phase (M/P) detectors; a signal generator coupled with the set of M/P detectors and the sensor array; and a computing device configured to control processes including: initiating: a) transmission of an excitation signal into the MUT with a transmitting electrode on the sensor array and b) transmission of a reference signal to the set of magnitude and phase (M/P) detectors; receiving a response signal from the MUT via a receiving electrode on the sensor array based on the excitation signal; separately comparing a magnitude and phase for each of the excitation signal and the reference signal with corresponding detection ranges for a first one of the M/P detectors; separately comparing a magnitude and phase for each of the response signal and the reference signal with corresponding detection ranges for a second one of the M/P detectors; iteratively adjusting the excitation signal until the response signal has both a magnitude and a phase within the corresponding detection ranges for the second M/P detector; and iteratively adjusting the reference signal until the reference signal has both a magnitude and a phase within the corresponding detection ranges for the first and the second M/P detectors.

(10) Particular aspects of the present subject matter provide electronic circuits, systems, and methods to apply an electronic circuit which: 1) generates an excitation signal and a reference signal at a specific frequency or over a range of frequencies; 2) applies the excitation a signal to a material under test (MUT) (which may include one or more subcomponents); 3) characterizes the response signal with respect to the reference signal; 4) determines the magnitude and phase relationship between the response signal produced in presence of the MUT relative to the reference signal; 5) computes the impedance and dielectric properties of the MUT (and in some cases, subcomponents); and 6) applies the measured dielectric properties to a physical model to correlate the measurement to a physical property or properties of the MUT (or a sample of the MUT that has been subjected to engineering testing to determine desired information about physical properties). The approaches described herein can include characterization methods for the measuring circuit board and sensor system, as well as a method to collect information in the form of electrical quantities with the circuit board and a sensor system.

(11) Various embodiments of the disclosure relate generally to an electronic circuit and system for the measurement of the impedance to electric current through sensing system in communication with a material under test (MUT) and subsequent extraction of the dielectric properties of the MUT. In some cases, the system includes a circuit having magnitude and phase detectors to measure the change in magnitude or strength and phase difference between a reference signal and an excitation (or transmit) signal and between the reference signal and the response (or received) signal produced

by the transit of the excitation signal through the MUT. The system can include at least one computing device configured to control the generation of the excitation and reference signals, to evaluate the measured signal levels, and to adjust excitation and/or reference signals to produce input signals to the magnitude and phase detectors and other circuit components that result in best performance of these detectors and components. Circuits according to various embodiments can include a signal strength determiner and/or phase determiner for determining the phase shift between the excitation signal, the reference signal, and the response signal specific to the MUT. The strength and phase determiner may be combined in a single circuit component. According to various embodiments, the measured difference in signal strength and phase are used to compute the (complex) electrical impedance and dielectric properties of the MUT. This (MUT/system-specific) impedance or the (MUT-specific) dielectric property can be correlated with a physical property or properties of the MUT. The system may be operated at a single frequency, or over a range of frequencies.

(12) In some particular embodiments, a system can include: a signal generator which generates the excitation signal and the reference signal; an excitation electrode connected with the signal generator and in electrically conductive or non-conductive contact with a material under test (MUT); a receiving electrode in electrically conductive or non-conductive contact with the material under test (MUT) which is part of the return path of the excitation current flowing from the excitation electrode through the MUT to the receiving electrode as the response signal; a reference signal which is compared to the excitation signal and the response signal at the receive electrode by a magnitude detector and/or phase detector; and at least one computing device connected with the signal generator, the signal strength determiner(s) and/or phase determiner(s) for the excitation and reference signals and for the response and reference signals, the at least one computing device configured to: send a control signal to the signal generator to initiate and conduct an excitation signal to the MUT via the excitation electrode and to the excitation-to-reference strength and phase determiner at a selected frequency or over a range of frequencies.

(13) Implementations may include one of the following features, or any combination thereof.

(14) In some particular embodiments, the at least one computing device receives digitized strength and phase data from analog to digital converters connected to the output of the strength and/or phase determiner(s) and communicates the data to another computing device to be used to compute the measured impedance or dielectric property, and to correlate impedance or dielectric property with a physical model of the MUT to quantify a physical property or properties of the MUT.

(15) In particular aspects, the computing device is further configured to: compute an electromagnetic property of the MUT based on the measured magnitude and phase for the response signal and the reference signal; and correlate the electromagnetic property with a physical property of the MUT based on a physical model of the MUT or a laboratory (or engineering) test of the MUT, wherein the electromagnetic property comprises one or more of: impedance, susceptance, permittivity or admittance.

(16) In some particular embodiments, the at least one computing device provides the computation of the measured impedance or the dielectric property to be correlated with a physical model of the MUT to quantify a physical property or properties of the MUT.

(17) In some particular embodiments, the at least one computing device provides the controlling function for the signal generator(s), switch(es), and other controllable elements of the circuit.

(18) In particular aspects, the first M/P detector provide a reading of the magnitude and phase of the excitation signal and the reference signal, and the second M/P detector provides a reading of the magnitude and phase of the reference signal and the response signal.

(19) In certain cases, the readings are obtained by a computing device configured to control the iterative adjustment (via the signal generator) of the amplitude and phase of the excitation signal and the reference signal.

(20) In some implementations, the reading for each of the excitation signal, the reference signal and

the response signal comprises separate magnitude and phase components.

(21) In particular cases, the iteratively adjusting includes adjusting an amplification of the excitation signal and/or the reference signal.

(22) In some aspects, a method further includes, after verifying that the excitation signal produces a response signal with a magnitude and phase within the corresponding detection ranges of the second M/P detector and a reference signal with a magnitude and phase within the corresponding detection ranges for both the first and the second M/P detectors: analyzing data obtained by the first and second M/P detectors using a data model about physical characteristics of the MUT to detect at least one physical characteristic of the MUT.

(23) In certain aspects, the analyzing includes correlating impedance or dielectric values for the MUT with an impedance value-to-physical characteristic correspondence table or a dielectric value-to-physical characteristic correspondence table. In some implementations, the correspondence table(s) are developed using a physical model of the MUT or by physical sampling or engineering evaluations of the MUT.

(24) In particular implementations, a method further includes converting the reference signal, the excitation signal and the response signal from analog format to digital format prior to separately comparing the reference signal, the excitation signal and the response signal with the corresponding detection ranges for the first and second M/P detectors.

(25) In some cases, the transmitting electrode includes a single transmitting electrode.

(26) In certain aspects, the receiving electrode includes a single receiving electrode that surrounds the transmitting electrode.

(27) In particular implementations, the receiving electrode includes a plurality of receiving electrodes, and the method further includes switching between the plurality of receiving electrodes for the response signal using an electrode switch.

(28) In some aspects, the excitation signal and the reference signal are both generated by a signal generator with a common control signal, and the excitation signal and the reference signal have a common frequency and a distinct magnitude and/or phase.

(29) Particular aspects of the present subject matter provide a system to generate electric signals at a specific frequency or over a range of frequencies and with varying levels of strength or magnitude to secure impedance or dielectric measurements on a Material Under Test (MUT) which then can be correlated with physical properties of the MUT. The system may include: at least one computing means which can receive parameters for a physical model of the MUT (and/or physical/engineering testing data about the MUT) and digitize data for the computation of the impedance or dielectric properties of the MUT; transmit control signals to the at least one signal generator and circuit components; and transmit data to a user interface; the at least one signal generator which generates two electric signals at a specific frequency or over a range of frequencies and at various amplitudes and phases of which: an excitation signal which is transmitted to an electrode in electrically conducting or non-conducting communication with the MUT and to the first of the at least two magnitude and phase detectors; and a reference signal which is transmitted to one of the at least two of the first magnitude and phase detector and the second of the at least two magnitude and phase detectors. The excitation signal which is transmitted to an electrode in communication with the MUT, produces a current through the MUT which is collected at least one receiving electrode which is in electrically conducting or non-conducting communication with the MUT and where the current is converted to a voltage (referred to as the received) signal that is transmitted to the second of the at least two magnitude and phase detectors. The magnitude and phase of the excitation signal relative to the reference signal from one of the at least two magnitude and phase detectors is transmitted as digital data to the at least one computing means. The magnitude and phase of the received signal relative to the reference signal from another of the at least two magnitude and phase detectors is transmitted as digital data to the at least one computing means. The at least one computing means processes the digitized magnitude and phase

data. The at least one computing means transmits the processed data to a user interface which communicates the desired physical properties of the MUT. The material under test may be a soil.

(30) Optionally, the material under test may be any material under test that produces a complex impedance spectrum when excited with methods of Electrical Impedance Spectroscopy (EIS). The specific frequency or range of frequencies applied may, in particular, fall within the range of 10 kHz to 100 MHz, and in some cases, 100 KHz to 100 MHz. In additional cases, a method comprises: at least one signal generator generating an electric excitation signal under control of an at least one computing means at a specific frequency or over a range of frequencies at specific amplitude within a range of amplitudes and a fixed phase whose voltage signal is transmitted to an electrode in communication with a material under (MUT) and produces an electric current through the MUT to a receiving electrode which is in electrically conducting or non-conducting communication with the MUT. The current collected at the receiving electrode is converted to a voltage signal (received signal) which is transmitted to one of the at least two magnitude and phase detectors where the at least one computing means determines if the amplitude of the received signal falls within the design amplitude input range of the magnitude and phase detector. If the amplitude of the received signal is not within the desired range for the magnitude and phase detector and the magnitude of the received signal relative to the reference signal is not within the desired tolerance band of the target magnitude, the computing means adjusts the amplitude of the excitation signal until the measured level is within the desired range and tolerance band. If the amplitude of the received signal is within the desired input amplitude range for the magnitude and phase detector and the magnitude of the received signal relative to the reference signal is within the tolerance band around the target magnitude, the excitation amplitude is fixed and the at least one computing means then instructs the at least one signal generator to generate electric signals at a fixed frequency or over a range of frequencies at a specific phase within a range of phases and at the fixed amplitudes. The electric signal with varying values of phase is transmitted to an electrode in electrically conductive or non-conductive communication with the MUT and produces a current through the MUT which is collected at a receiving electrode which is in electrically conductive or non-conductive communication with the MUT. The current collected at the receiving electrode is converted to a voltage signal (received signal) which is transmitted to one of the at least two magnitude and phase detectors where the at least one computing means determines if the phase of the received signal falls within the design input phase range of the amplitude and phase detector. If the phase of the received signal is not within the desired range for the magnitude and phase detector is not within the desired tolerance band around the desired target phase output, the computing means adjusts the phase of the excitation signal until the phase of the received signal is within the desired range and within the desired tolerance band. If the phase of the received signal is within the desired phase input range for the level and phase detector and within the desired tolerance band, the amplitude and phase are fixed and used for the values of the electric excitation signal which is measured by the other of the at least two magnitude and phase detectors. The at least one computing means then directs the at least one signal generator to generate a reference signal whose amplitude and phase fall within the design ranges of the one of the at least two magnitude and phase detectors used to measure the received signal and is adjusted to produce the magnitude and phase outputs within the desired tolerance band about the target magnitude and phase of the one of the at least two magnitude and phase detectors used to measure the received signal. The at least one computing means receives and processes the digitized magnitude and phase measurement from the at least two magnitude and phase detectors for the excitation signal and the received signal. The at least one computing means transmits the processed data to a user interface which communicates the desired physical properties of the MUT.

(31) In certain cases, the physical model further defines amplitudes of an excitation signal and the reference signal for correlating the measurement with the at least one physical property of the MUT.

(32) Various approaches can be used to determine electromagnetic properties of the MUT, including, e.g., one or more of: impedance, susceptance, permittivity or admittance.

(33) Two or more features described in this disclosure, including those described in this summary section, may be combined to form implementations not specifically described herein. The system and in particular its computing device may be configured to carry out the methods described herein. Further variants of the described methods result from the intended use of the described system and its components.

(34) The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, objects and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1. shows an illustration of a phase and magnitude spectrum over the range of frequencies of interest for a typical soil tested with a typical embodiment of an impedance measurement system.

(2) FIG. 2 Shows and illustration of the Keysight E4990A Impedance Analyzer.

(3) FIG. 3 shows a general configuration of a sensor system including an impedance measurement circuit according to various embodiments of the disclosure.

(4) FIG. 4 shows an alternate configuration of a sensor system including an impedance measurement circuit with selected circuit elements according to various embodiments of the disclosure

(5) FIG. 5. shows an alternate configuration of a sensor system including an impedance measurement circuit with selected circuit elements according to various embodiments of the disclosure and an alternate sensor configuration.

(6) FIG. 6 is a flow diagram illustrating processes according to various implementations.

(7) FIG. 7 is a flow diagram illustrating processes according to various additional implementations.

(8) It is noted that the drawings of the various implementations are not necessarily to scale. The drawings are intended to depict only typical aspects of the disclosure, and therefore should not be considered as limiting the scope of the implementations. In the drawings, like numbering represents like elements between the drawings.

DETAILED DESCRIPTION

(9) The various circuits, systems, methods, and procedures described herein are related to the generation of electric signals at a single frequency or over a range of frequencies for measuring the impedance or dielectric properties of a material under test (MUT). The frequency or range of frequencies is selected on the basis of the properties of the MUT for an impedance-spectroscopic analysis for the determination of physical properties of the MUT. In certain implementations, a single frequency is adequate for tomographic analysis, e.g., as described in US Patent Publication Nos. 2016/0161624, and 2018/0128934, and U.S. Pat. Nos. 9,465,061, 9,804,112, and 10,324,052; or for the determination of physical properties of selected materials that act as pure capacitors, such as hot mix asphalt, as described in U.S. Pat. Nos. 5,900,736, 6,414,497, and 6,803,771. However, for a more complex MUT such as a soil, an impedance-spectroscopic analysis over a range of frequencies is required, e.g., as described in U.S. Pat. Nos. 7,219,024, 9,465,061, 9,805,146, and 10,161,893. Each of the afore-mentioned applications, publications and issued patents is hereby incorporated by reference in its entirety.

(10) While certain example implementations are described with reference to determining electromagnetic properties of an MUT such as impedance, various approaches can be used to determine additional, or alternative, electromagnetic properties of the MUT, including, e.g., one or

more of: susceptibility, permittivity or admittance.

(11) Commonly labeled components in the FIGURES are considered to be substantially equivalent components for the purposes of illustration, and redundant discussion of those components is omitted for clarity.

(12) While one example MUT described in this disclosure is a soil, the various circuits, systems, methods and procedures described herein are applicable to any material under test that has a complex impedance spectrum, e.g., where Electrical Impedance Spectroscopy (EIS) may be applied. For example, U.S. Pat. No. 9,465,061 describes a method of conducting an in-process inspection of solid materials with EIS. In certain cases, there is a need to conduct in-process inspections and characterizations of fluids, as suggested by U.S. Pat. Nos. 9,372,183, 9,389,175, and 9,797,855. U.S. Pat. No. 9,389,175 applies an optical detection system, and U.S. Pat. Nos. 9,372,183 and 9,797,855 apply impedance flow cytometry, which counts and characterizes cells. Each of the afore-mentioned US patents is incorporated by reference in its entirety. Additional publications discuss various electromagnetic methods of characterizing dairy products (e.g., milk) and other foods such as olive oil, fruits, vegetable oils, cookies, pork, and fish. In all these examples, the complex impedance spectrum extends over a broad range of frequencies. This is shown in FIG. 1, illustrating corresponding Phase and Magnitude graphs over a frequency range for a typical soil tested with a conventional impedance soil measurement system. This broad range of frequencies makes it difficult to secure accurate measurements that can be used to correlate the measured impedance spectrum with a physical property or properties of the MUT, for example, density (or compaction level) and/or moisture level for a soil.

(13) A typical range of frequencies of interest for soils is from about 10 kHz to approximately 100 MHz, and in particular cases, from about 100 kHz to about 100 MHz. These ranges of frequencies have been shown to provide an impedance spectrum that can be used to correlate with physical properties such as density (compaction level) and moisture level of the soil. As can be seen in FIG. 1, the values of the phase (e.g., about 45 degrees up to about 70 degrees) and magnitude (e.g., about -5 DB to about 5 DB) over the range of frequencies is very large. These values are used to compute the complex impedance of the MUT over the frequency range of interest. In order to secure accurate data over the entire range of frequencies, a very sophisticated (and expensive) instrument is typically required. One such instrument is the Keysight Technologies E4990A Impedance Analyzer, an image of which is shown in FIG. 2. This Impedance Analyzer is a 30-pound (14 kg) instrument that is about 17.0-in wide, 9.3-in high and 11.7-in deep (432 mm×239 mm×296 mm) and requires laboratory environmental conditions to function effectively. As such, the conventional systems and approaches are unwieldy in practical, field use scenarios. To address these shortcomings in conventional systems and approaches, various disclosed implementations provide circuits, systems, and methods for use in the field and/or production environment that are able to secure comparable levels of data accuracy from an MUT over the desired frequency range. Other approaches to satisfying this objective are presented in US Patent Publication No. 2014/0266268 and U.S. Pat. No. 10,330,616 (each incorporated by reference in its entirety).

(14) While certain frequency ranges of interest are described herein, other ranges and sub-ranges may also be of interest. For example, the systems and approaches described herein can be applicable to investigating soils in frequencies around 32 kHz, e.g., 32 kHz±10 kHz, 15 kHz or 20 KHz.

(15) Various commercially available electronic components are described as examples relating to the various embodiments of the disclosure. These are used only for illustrative purposes. Other such components may be used as selected by one skilled in the art.

(16) Referring to FIG. 3, a schematic depiction of a measurement system (or simply, system) **100** is shown according to various implementations. In some cases, the measurement system **100** interacts with (or includes) a plurality of elements (or, sub-systems). In various implementations, system **100** includes a circuit to generate electric signals at a specific frequency (or over a range of

frequencies), along with controls (e.g., control software such as program code) to communicate with the other elements of an overall measurement system. As shown in FIGURE (FIG. 3, the measurement system **100** is configured to interact with a sensor system **200**, which in turn communicates with a MUT **210**. System **100** is configured to receive a physical model **300**, which in some example cases is the physical model of the MUT **210**, e.g., soil. That is, the physical model **300** can include a physical soil model where the MUT **210** includes soil. Additionally, the sensor system **200** can communicate with an interface (e.g., user interface) **400**, which can be local or remote relative to the system **100**. In some cases, the interface **400** enables a user or other operator to communicate desired physical attributes for testing of the MUT, e.g., for soil this could include density and/or moisture.

(17) In various implementations, the sensor system **200** includes a central excitation electrode (TX **201**) surrounded by a concentric coplanar ring including a receiving electrode (RX **202**). In some cases, the design of the sensor is described in U.S. Pat. Nos. 5,900,736 and 7,219,024. In some cases, the electrodes **201**, **202** are in direct physical contact with the MUT **210** (e.g., as described with respect to the sensor system in U.S. Pat. No. 5,900,736). In other cases, the electrodes **201**, **202** are offset (separated) from the MUT **210** (e.g., such that a gap or standoff is present between the electrodes **201**, **202** and the MUT **210**, as is described with respect to the sensor system in U.S. Pat. No. 7,219,024). The sensor system **200** and MUT **210** present a resistive and/or capacitive load on the circuit of the measurement system **100**. The system **100** is configured to account for and accommodate for the load characteristics in the sensor system **200** and MUT **210** in order to secure accurate readings of the change in the magnitude and phase of the response signal (received at RX **202**) compared to the excitation signal (sent via TX **201**).

(18) The physical model of the MUT (e.g., soil) **300** is used in setting the parameters of the circuit in system **100**. U.S. Provisional Patent Application No. 62/661,682 describes a system and method for securing a physical model of soils for use with system **100**.

(19) Certain implementations may replace the physical model and/or supplement the physical model of the MUT **300** with testing data about physical characteristics of the MUT **300**, e.g., an ASTM test determining physical characteristics of the MUT **300**.

(20) Interface **400** can include any conventional interface, e.g., a user interface, to enable a user such as a human user or other communication system to communicate physical properties of the MUT **210** from the measurement system **100**, along with enabling data logging such as data about time and/or location of the measurement. Example interfaces can include a laptop computer, a tablet, a smart phone, or a dedicated user interface that is connected (e.g., physically and/or wirelessly) with the measurement system **100**. In any case, the interface **400** is configured to communicate data from the measurement system **100** in real time, e.g., in a field-testing environment.

(21) As shown in FIG. 3, in various implementations, system **100** can include at least one computing device (e.g., processor and/or memory) **10** that is configured to receive data from external systems, e.g., to receive the physical (MUT) model **300**, as well as send and receive data to/from the interface system **400**. In some cases, the physical (MUT) model **300** includes information about a desired frequency range and expected values of the impedance spectrum for a response from the MUT **210**. In certain aspects, information about the desired frequency range needed and the possible range of measured magnitudes and phases of the impedance spectrum are obtained from prior field testing or laboratory testing with a system as described in U.S. Provisional Patent Application No. 62/661,682. The computing device **10** sends a control signal **11** to a signal generator **12** and receives digital data **18**, **19** about the magnitude (level) and phase of the generated signal from respective magnitude and phase (M/P) detectors (e.g., TX and RX) **15** and **17**. The (TX) M/P detector **15** compares the excitation TX signal **13** with a reference signal **14**, both generated by the signal generator **12**. The (RX) M/P detector **17** compares the reference signal **14** with the receive (response) RX signal **16** (received at electrode RX **202**). The excitation voltage

signal **13** causes current flow from the TX electrode **201** on the sensor, through the MUT **210**, to the receiving electrode RX **202**, where it is measured as the voltage (response) signal **16** at the RX level and phase detector **17**. The (digital) output signals from the magnitude and phase (M/P) detectors **15** and **17** are transmitted to the computing device **10**. The computing device **10** includes a processor for processing the magnitude and phase data, and a communication system (e.g., conventional wireless and/or wireless communication system) for sending the results as an output, e.g., at interface **400**.

(22) FIG. **4** is a schematic depiction of a system **500** according to various additional implementations. Commonly labeled features within the FIGURES can represent the same or similar components, separate descriptions of which may be omitted. In some cases, the system **500** includes at least one computing device **510**, which can include at least one processor and/or a memory. In certain implementations, the computing device **510** includes a digital signal controller, e.g., similar to a dsPIC33F 16-bit Digital Signal Controller. However, other microprocessors may be used in additional implementations. In particular implementations, system **500** further includes a signal generator **104**, such as a dual channel Direct Digital Synthesis (DDS) chip. The signal generator **104** is connected to the computing device **510** and is configured to receive control instructions as described herein. In particular examples, the signal generator **104** includes a device such as an Analog Devices AD9958. In some cases, multiple single-channel signal generator chips may be used in place of, or in addition to a dual-channel ship. The signal generator **104** is configured to generate two signals at the same frequency at different levels (magnitudes, or strengths): an excitation (TX) signal **107**, and a reference signal **114**. In various implementations, the amplitudes (or, magnitudes) of the signals, as well as the phases, are separately set at different values for the excitation signal **107** and the reference signal **114**. In certain implementations, system **500** includes an amplifier **105** for amplifying the excitation signal **107** and transmitting an amplified (TX) signal **106** to the transmit electrode **201** of the sensor system **200** and to a (TX) magnitude and phase (M/P) detector **109**. An example amplifier **105** can include an AD4870 (from Analog Devices) or a similar device. In certain implementations, the computing device **501** sends an amplifier control signal **103** to control the amplifier **105**, e.g., in amplifying excitation signal **107**. In the example embodiment of system **500** shown in FIG. **4**, an additional magnitude and phase (M/P) detector **115** is shown for detecting the response signal **121** from the response (receiving) electrode **202**. In some cases, the M/P detector **115** is a similar model/make as M/P detector **109**, however, in other cases, these M/P detectors are distinct types. In some cases, the M/P detector **115** is an AD8302 or similar device. In practice, the computing device **510** instructs the signal generator **104** to generate the reference signal **114** and the excitation (TX) signal **107**. The reference signal **114** is transmitted to the (TX) M/P detector **109** and also to the response M/P signal detector (RX) **115**. The excitation voltage signal **107**, once amplified and transmitted as TX signal **106**, produces current flow from the TX electrode **201** on the sensor system **200**, through the MUT **210**, to the receiving electrode RX **202**. The response signal **121** (in response to the excitation) is measured at the response M/P detector (RX) **115**. In certain cases, the two M/P detectors **109**, **115** (e.g., including an AD8302) produce analog output signals proportional to the logarithmic amplitude ratio (also referred to as magnitude) of and the phase difference between, respectively, the (TX) excitation signal **106** and the reference signal **114** (in M/P detector **109**), and the (RX) receive signal **121** and the reference signal **114** (in M/P detector **115**). In various implementations, these two magnitude and phase analog signals **110**, **111**, **116**, **117**, one set for each of magnitude and phase detectors (TX **109** and RX **115**), are converted to digital signals by an analog to digital converter (ADC) **114** prior to being transmitted back to the computing device **510** (e.g., for storage, processing, output to a user interface **400**, etc.). In this example embodiment of system **500**, the analog to digital converter (ADC) **114** can include Maxim MAX **1239** analog-to digital converter or similar device, that accepts the two sets of analog magnitude and phase signals and converts them to digital signals for transmission to the computing device **510**.

(23) As noted herein, the amplitude of the excitation signal **107** generated with the signal generator **104** may be amplified by the amplifier **105**. In some cases, after passing through the amplifier **105**, the TX (voltage) signal **106** is measured with the M/P detector **109** and also applied to the TX (excitation) electrode **201** on the sensor system **200**. This TX signal **106** causes a current flow through the MUT **210**, which is detected at the receive electrode (RX) **202**, where it is transmitted as the measurable voltage (RX) signal **121** to (RX) M/P detector **115**. The amplitude of the receive (RX) signal **121** is dictated by the amplitude of the (TX) signal **106** and the dielectric properties of the MUT **210**. By changing the amplitude of the TX signal **106** in either the signal generator **104** or the amplifier **105** (e.g., via instructions from computing device **510**), the amplitude of the receive (RX) signal **121** measured by the M/P detector **115** is changed. In various implementations, the computing device **510** is configured (e.g., programmed) to adjust the amplitude of the output (TX) signal **106** (e.g., via adjustment to the DDS control signal **102** and/or the amplifier control signal **103**) until the receive (RX) signal **121** detected at the M/P detector **115** falls within one or more predefined ranges (e.g., an input range for the M/P detector **115**, as well as a distinct, but overlapping input range for the M/P detector **109**). In some cases, adjusting the receive (RX) signal **121** to fall within one or more range(s) of the (RX) M/P detector **115** may necessitate a (TX) signal **106** that would fall outside the range of the (TX) M/P detector **109**. In these cases, additional measures of proportionate attenuation of the stronger (TX) signal **106** would be required, e.g., by an attenuator before M/P detector **109** (not shown). In various implementations, the phase of the excitation (TX) signal **107** from the signal generator **104** is independently adjusted by the DDS Control signal **102**, in order to adjust the phase of the receive (RX) signal **121**. In other implementations, e.g., where the M/P detector **115** has a broad acceptable phase range (e.g., such as where the M/P detector **115** includes an AD8302 chip), the computing device **510** need not independently adjust the phase of the excitation (TX) signal **107**. In particular cases, the amplitude and phase of the reference signal **114** is specifically defined for TX and RX measurements within the input range specified for the M/P detectors (e.g., Analog Devices input specifications for the AD8302 chip).

(24) In various implementations, the ability to independently set the amplitude of the excitation and reference signals to enhance performance and compatibility of measured signals with the signal input range of the magnitude and phase detectors significantly improves the accuracy of data about the MUT **210**, as compared with conventional approaches. In the illustration of system **500** (FIG. 4), the function of magnitude and phase detection is performed by M/P detectors **109**, **115** (e.g., Analog Devices AD8302 or other comparable magnitude and phase detectors). Having two comparisons against the reference signal (excitation (TX) **106** vs. reference **114** and receive (RX) **121** vs. reference **114**) enables the computing device **510** to compute the complex ratio of excitation (TX) voltage signals and response (RX) voltage signals, expressed in terms of magnitude (e.g., logarithmic amplitude ratio) and phase difference as:

(25) $M_{TX-REF} - M_{RX-REF} = M_{TX-RX} P_{TX-REF} - P_{RX-REF} = P_{TX-RX}$ where $M_{sub.TX-REF}$ is the magnitude of the TX signal **107** relative to the reference signal **114**; where $M_{sub.RX-REF}$ is the magnitude of the RX signal **121** relative to the reference signal **114**; where $M_{sub.TX-RX}$ is the magnitude of the TX signal **107** relative to the RX signal **121**; where $P_{sub.TX-REF}$ is the phase of the TX signal **107** relative to the reference signal **114**; where $P_{sub.RX-REF}$ is the phase of the RX signal **121** relative to the reference signal **114**; and where $P_{sub.TX-RX}$ is the phase of the TX signal **107** relative to the RX signal **121**.

(26) After calculating the magnitude and phase differences between the excitation **107** and response **121** signals, the computing device **510** is configured to compare those magnitude and phase differences with the physical model of the MUT (e.g., soil) **300** to determine physical properties of the MUT. As noted herein the physical model of the MUT **300** can include magnitude and phase correspondence information (e.g., tables, correlations, etc.) with physical properties of an MUT. For example, for particular MUT types (e.g., soil), the physical model **300** includes magnitude

and/or phase ranges for signal responses that correspond with particular physical properties or characteristics of the MUT. In particular examples, density or moisture content values or ranges are correlated with distinct magnitude and/or phase values or ranges, such as those calculated by the computing device **510** using the approaches described herein.

(27) The measured magnitudes M.sub.TX-REF and M.sub.RX-REF and phases P.sub.TX-REF and P.sub.RX-REF can also be used to characterize the measurement systems described herein (e.g., measurement system **100** in FIG. 3, measurement system **500** in FIG. 4, and measurement system **600** in FIG. 5) when replacing an unknown MUT with an object of known electrical properties. For example, in combination with information regarding components of the measurement systems (from additional independent measurements and/or component specifications), the measurement of materials of known electrical properties, such as polyethylene, glass, G9 composite, and graphite, enables calibration and establishment of a model for conversion of impedance data to dielectric properties.

(28) FIG. 5 shows an additional system **600** according to various implementations. System **600** can include a number of components described with respect to other systems disclosed herein, e.g., measurement systems **100** (FIG. 3) and **500** (FIG. 4). In certain implementations, system **600** includes the measurement system **500** shown and described with reference to FIG. 4. However, in contrast to the depiction in FIG. 4, system **600** includes a distinct sensor system **200A** that includes an RX switch **205** in communication with computing device **510**. In certain implementations, the electrodes in sensor system **200A** also differ from the electrodes in sensor system **200** (FIG. 4). That is, electrodes **201**, **202**, **203** in sensor system **200A** are arranged in a linear (or semi-circular) array of sensors. This arrangement of the sensors permits an examination of the spatial or tomographic distribution of physical properties of the MUT **210**. In this example implementation, the sensor system **200A** includes one excitation electrode (TX) **201** and two distinct receive (RX) electrodes **202** and **203**. A similar electrode arrangement is illustrated in US Patent Publication No. 2018/0128934 and U.S. Pat. Nos. 9,465,061 and 9,804,112 (all incorporated by reference herein). In various implementations, the excitation electrode (TX) **201** is always active while one active receive electrode (selected from the receive (RX) electrodes **202** or **203**), is selected via a control signal **130** from the computing device **510** and the RX switch **205**. The volumes of the MUT that are measured differ between the measurement with transmitting electrode (TX) **201** and receiving electrode (RX) **202** versus measurement with transmitting electrode (TX) **201** and receiving electrode (RX) **203**. Applying the teachings of U.S. Pat. Nos. 9,465,061 and 9,804,112, the impedance spectrum of different volumes of the MUT **210** may be determined and correlated to physical properties of those volumes

(29) In various implementations, the computing devices (e.g., computing device **10**, FIG. 3, or computing device **510**, FIGS. 4 and 5) can be configured to adjust the TX signal and RX signal to improve the accuracy of detecting one or more physical characteristics of the MUT **210**. In particular cases, the computing devices disclosed herein are configured to match the input excitation signal (e.g., excitation (TX) signal **13** or excitation (TX) signal **106**) and receive signal (e.g., receive (RX) signal **16** or receive (RX) signal **121**) to the design range of the M/P detectors (e.g., M/P detectors **15**, **17** in FIGS. 3 and M/P detectors **109**, **115** in FIGS. 4 and 5), as well as adjust the reference signals (e.g., reference (REF) signal **14** in FIG. 3 and (REF) signal **114** in FIGS. 4 and 5) to obtain precise and accurate data from the MUT **210** by keeping the (TX) signal **106**, the (RX) signal, and the (REF) signal **114** all within the input specifications for the (TX) M/P detector **109** and the (RX) M/P detector **115**.

(30) FIG. 6 is a flow diagram illustrating processes in a method according to various implementations. In some cases, the method includes characterizing an excitation signal, for example, to detect physical characteristics of an MUT. In particular cases, the method also includes detecting (or characterizing) at least one physical characteristic of the MUT using the excitation signal(s). As will be evident, processes described with reference to FIG. 6 can be applied to any of

the systems disclosed herein, e.g., measurement system **100**, measurement system **500** and/or measurement system **600**. Turning to FIG. **6**, processes can include:

(31) Process P1: transmitting an excitation signal (e.g., excitation signal **13** or excitation signal **107**) into the MUT **210** using a transmitting electrode (e.g., TX **201**) on a sensor array (e.g., sensor system **200**, **200A**) and transmitting a reference signal (e.g., REF **14** or REF **114**) to a set of magnitude and phase (M/P) detectors (e.g., M/P detectors **15**, **17** or M/P detectors **109**, **115**). In certain cases, as noted herein, the excitation signal (e.g., excitation signal **13** or excitation signal **107**) and the reference signal (e.g., REF **14** or REF **114**) are both generated by a signal generator (e.g., signal generator **12** or signal generator **104**) with a common control signal (e.g., control **11**, control **102**), where the excitation signal and the reference signal have a common frequency and a distinct magnitude and/or phase.

(32) Process P2: receiving a response signal (e.g., receive signal **16** or receive signal **121**) from the MUT **210** via a receiving electrode (e.g., RX **202** or RX **203**) on the sensor array (e.g., sensor system **200**, **200A**) based on the excitation signal (e.g., excitation signal **13** or excitation signal **107**).

(33) Decision D1: separately comparing a magnitude and phase for the response signal (e.g., response signal **16** or response signal **121**) and the reference signal (e.g., REF **14** or REF **114**) with corresponding detection ranges for a second one of the M/P detectors (e.g., M/P detector **17** or M/P detector **115**). As described herein, the second M/P detector (e.g., M/P detector **17** or M/P detector **115**) provides a reading (e.g., to computing device **10** or **510**) of the magnitude and phase of the reference signal (e.g., REF **14** or REF **114**) and the response signal (e.g., response signal **16** or response signal **121**). The reading for each of the reference signal (e.g., REF **14** or REF **114**) and the response signal (e.g., response signal **16** or response signal **121**) includes separate magnitude and phase components (e.g., FIGS. **4** and **5**).

(34) If No to Decision D1, process P3 includes adjusting the excitation signal (e.g., excitation signal **13** or excitation signal **107**), e.g., by increasing or decreasing the amplitude or magnitude of that signal, e.g., via a control signal **102** to the signal generator (e.g., signal generator **13** or signal generator **104**). In certain implementations, as illustrated in the embodiments in FIGS. **3** and **4**, the computing device (e.g., computing device **10** or computing device **510**) adjusts the amplification of the excitation signal (e.g., excitation signal **107**), e.g., generating TX signal **106**.

(35) If Yes to Decision D1, Decision D2 includes separately comparing a magnitude and phase for the reference signal (e.g., REF **14** or REF **114**) with corresponding detection ranges for each of a first one of the M/P detectors (e.g., M/P detector **15** or M/P detector **109**) and a second one of the M/P detectors (e.g., M/P detector **17** or M/P detector **115**). As described herein, the first M/P detector (e.g., M/P detector **15** or M/P detector **109**) provides a reading (e.g., to computing device **10** or **510**) of the magnitude and phase of the excitation signal and the reference signal, and the second M/P detector (e.g., M/P detector **17** or M/P detector **115**) provides a reading (e.g., to computing device **10** or **510**) of the reference signal and the response signal. In some cases, the reading for each of the received excitation signal, the reference signal and the response signal includes separate magnitude and phase components (e.g., FIGS. **4** and **5**).

(36) If No to Decision D2 (meaning that reference signal deviates from range of both the first and second M/P detectors), proceed to process P4, which includes adjusting the magnitude and/or phase of the reference signal (e.g., REF **14** or REF **114**). As with the decision loop D1, negative responses to decision loop D2 can trigger iterative adjustment of the reference signal until both the response signal requirements (Decision D1) and the reference signal requirements (Decision D2) are satisfied.

(37) A Yes response to Decision D2 means that the response signal (e.g., receive signal **16** or receive signal **121**) and the reference signal (e.g., REF **14** or REF **114**) have both a magnitude and a phase within the corresponding detection ranges for the second M/P detector (e.g., M/P detector **17** or M/P detector **115**), and the reference signal (e.g., REF **14** or REF **114**) has a magnitude and

phase with the corresponding detection ranges for both the first and second M/P detectors (e.g., M/P detectors **15** and **17** or M/P detectors **109** and **115**). In this case, process P5 can include analyzing the response signal data from the second M/P detector (e.g., M/P detector **17** or M/P detector **115**) and the reference signal (REF **14** or REF **114**) using a data model (e.g., physical model **300**) about physical characteristics of the MUT **210** to detect at least one physical characteristic of the MUT **210**. In particular cases, this analysis includes correlating impedance or dielectric values in the response signal (e.g., receive signal **16** or receive signal **121**) with a response signal-to-physical characteristic correspondence table (e.g., in physical model **300**).

(38) One or more of the above-noted processes can be modified according to the various implementations described herein. For example, signals such as the reference signal, the excitation signal and the response signal can be converted from analog format to digital format prior to separately comparing those signals with the corresponding detection ranges for the first and second M/P detectors **109**, **115**, e.g., as illustrated in FIGS. **4** and **5**. Even further, in cases where the sensor system (e.g., sensor system **200A**) includes a plurality of distinct receiving electrodes (e.g., RX **202**, RX **203**), approaches can include switching between receiving electrodes (e.g., RX **202** or RX **203**) for the response signal using an electrode switch (e.g., RX switch **205**, FIG. **5**).

(39) FIG. **7** is a flow diagram illustrating processes in an additional method of detecting a physical characteristic of an MUT **210** using one or more systems herein. In various implementations, the method is performed by execution of program code or other programmable instructions at a processor such as the processor(s) in the computing device **10** of FIG. **3** or the computing device **510** of FIGS. **4** and **5**.

(40) In particular implementations process P101 includes obtaining a physical model of the MUT **300**, e.g., as a data file or set of data files, via one or more communications media. The physical model of the MUT **300** includes the frequency range over which impedance and/or dielectric data must be secured from the MUT **210** in order to correlate that impedance and/or dielectric data with one or more physical properties of the MUT **210**. The model **300** can also include information about the amplitudes of the excitation signal and reference signal for collecting data about the MUT **210**.

(41) In process P102, the computing device (e.g., computing device **10** or computing device **510**) transmits a control signal (e.g., control signal **11** in FIG. **1** or control signal **102** in FIGS. **4** and **5**) to a signal generator (e.g., signal generator **12** or **104**) to initiate generation of excitation and reference signals (e.g., signals **13** and **14**, or signals **107** and **114**) at a frequency and amplitude (or over a range of frequencies and amplitudes) within a predefined range(s) of amplitudes and phase for the M/P detectors (e.g., M/P detectors **15**, **17** or M/P detectors **109**, **115**). In certain implementations, information about the desired frequency range and the possible range of measured magnitudes and phases of the impedance spectrum is obtained from prior field testing or laboratory testing with a system as described in U.S. Provisional Patent Application No. 62/661,682, previously incorporated by reference herein. In the case of measurement system **500** (FIGS. **4** and **5**), the computing device can also be configured to send an amplifier control signal **103** to the amplifier **105** for amplifying the excitation signal **107**.

(42) Also in P102, in response to the control signals from the computing device, the signal generator (e.g., signal generator **12** or **104**) generates an excitation signal **13** (FIG. **3**) or **107** (FIGS. **4** and **5**), and in particular cases, the amplifier **105** (FIGS. **4** and **5**) amplifies the excitation signal, according to the commanded frequency, amplitude and phase dictated by the control signals. The TX signal (e.g., TX signal **13**, FIG. **3**, or TX signal **106**, FIGS. **4** and **5**) is applied to the excitation electrode **201** in the sensor system **200** (FIGS. **3** and **4**) or **200A** (FIG. **5**). The current produced by the excitation signal passes through the MUT **210** and is detected at the receiving electrode **202** (FIGS. **3** and **4**) or multiple receiving electrodes **202**, **203** (FIG. **5**). The detected current at receiving electrode(s) is converted to the (received) voltage signal (e.g., RX signal **16** in FIG. **3** or RX signal **121** in FIGS. **4** and **5**), which is measured in M/P detector(s) (e.g., M/P detector **17** in

FIG. 3 or M/P detector **115** in FIGS. 4 and 5). The M/P detector produces magnitude and phase data about the RX signal. In the illustrations in FIGS. 4 and 5, the magnitude and phase detector is configured to produce analog output voltages. In these cases, the voltages, which are proportional to the measured magnitude and phase, are converted to digital signals in the analog to digital converter **114** and transmitted to the computing device **510 101**. In FIG. 3, the M/P detector **17** transmits the signals to the computing device **10**.

(43) In P103, the amplitude of the receive signal (e.g., receive signal **16** in FIG. 3 or receive signal **121** in FIGS. 4 and 5) is compared to the specified amplitude input range of the M/P detector (e.g., M/P detector **17** in FIG. 3 or M/P detector **115** in FIGS. 4 and 5). If the measured magnitude, which quantifies the amplitude of the receive signal (e.g., receive signal **16** in FIG. 3 or receive signal **121** in FIGS. 4 and 5) relative to the commanded amplitude of the reference signal (e.g., REF signal **14** in FIG. 3 or REF signal **114** in FIGS. 4 and 5), indicates that the amplitude of the receive signal is not within the specified range, P101 and P102 are repeated with a different amplitude of the excitation signal (**106**). In P104, the excitation amplitude is adjusted to bring the amplitude of the receive signal (e.g., receive signal **16** in FIG. 3 or receive signal **121** in FIGS. 4 and 5) within the specified input range of the M/P detector (e.g., M/P detector **17** in FIG. 3 or M/P detector **115** in FIGS. 4 and 5), for example, to produce a desired quality of the magnitude output signal. In some cases, this includes keeping the measured magnitude of the (TX) signal **106**, the (RX) signal, and the (REF) signal **114** all within the input specifications for the (TX) M/P **109** and (RX) M/P **115**. Otherwise, the procedure moves to process P105 if the measured phase of the (TX) signal **106** or the (RX) signal are outside of the specified phase range limitations for the M/P detector **115**, or to process P107 if phase adjustments are not necessary.

(44) In P107, the computing device (e.g., computing device **10** or computing device **510**) adjusts the phase of the signal generated by the signal generator (e.g., signal generator **12** or **104**) to match the phase of the signal required for the specified range of the M/P detector (e.g., M/P detector **17**, FIG. 3 or M/P detector **115**, FIGS. 4 and 5). The data from the last reading from process P105 provides the initial value of phase for process P106.

(45) In P106, the measured receive signal (e.g., RX signal **16**, FIG. 3, or RX signal **121**, FIGS. 4 and 5) is compared to the specified input phase range for the magnitude and phase detector (e.g., M/P detector **17**, FIG. 3 or M/P detector **115**, FIGS. 4 and 5). If the measured phase is not within the specified input range of the RX M/P **115** or the TX M/P **109**, P106 is repeated with a different phase of the excitation signal (e.g., TX signal **13**, FIG. 3 or TX signal **106**, FIGS. 4 and 5) with the magnitude of the excitation signal fixed. In P108, the phase is adjusted to bring the phase of the receive signal (e.g., RX signal **16**, FIG. 3, or RX signal **121**, FIGS. 4 and 5) within the specified input range of the phase detector (e.g., M/P detector **17**, FIG. 3 or M/P detector **115**, FIGS. 4 and 5). Otherwise, the process moves to Process P109.

(46) At this point, the frequency, amplitude and phase of the excitation signal (e.g., TX signal **13**, FIG. 3 or TX signal **106**, FIGS. 4 and 5) and receive signal (e.g., RX signal **16**, FIG. 3, or RX signal **121**, FIGS. 4 and 5) are fixed. In P109, the reference signal (e.g., REF **14**, FIG. 3 or REF **114**, FIGS. 4 and 5) is adjusted for the receive (RX) magnitude and phase measurement. The amplitude and phase of this signal is selected to be within the respective specified input ranges of the magnitude and phase detector **115** (e.g. the specifications of the Analog Devices AD8302) such that the amplitude of the reference signal is as close as possible to the amplitude of the receive (RX) signal and the phase difference is 90°, where the phase difference can be detected with highest precision.

(47) In P110, the measured magnitude and phase of the receive signal (e.g., RX signal **16**, FIG. 3, or RX signal **121**, FIGS. 4 and 5) are compared to target output magnitude and phase of the level and phase detector (e.g., M/P detector **17**, FIG. 3 or M/P detector **115**, FIGS. 4 and 5). If the measured magnitude and phase from detector (e.g., M/P detector **17**, FIG. 3 or M/P detector **115**, FIGS. 4 and 5) is not within specified tolerance bands based on the signal to noise ratios around the

target magnitude and phase, P109 is repeated with a magnitude and phase of the reference signal (e.g., REF **14**, FIG. **3** or REF **114**, FIGS. **4** and **5**) that is increased or decreased to match the target values. Otherwise, the procedure is continued in P110.

(48) In P110, the reference signal (e.g., REF **14**, FIG. **3** or REF **114**, FIGS. **4** and **5**) is adjusted for the excitation (TX) magnitude and phase measurement. The amplitude and phase of this signal is selected to be within the respective specified input ranges of the magnitude and phase detector (e.g., M/P detector **15**, FIG. **3** or M/P detector **109**, FIGS. **4** and **5**) such that the amplitude of the reference signal is approximately equal to the amplitude of the TX signal and the phase difference is 90° , where the phase difference can be detected with sufficient precision.

(49) In P112, the measured magnitude and phase of the excitation signal (e.g., TX signal **13**, FIG. **3** or TX signal **106**, FIGS. **4** and **5**) are compared to target output magnitude and phase of the level and phase detector (e.g., M/P detector **15**, FIG. **3** or M/P detector **109**, FIGS. **4** and **5**). If the measured magnitude and phase from detector **109** is not within specified tolerance bands based on the signal to noise ratios around the target magnitude and phase, P111 is repeated with different amplitude and phase of the reference signal (**114**). Otherwise, the procedure is repeated with a different frequency within the specified range of frequencies starting in Process P102 or completed in Process P112.

(50) The computation of the magnitude and phase of the excitation signal **106** relative to the receive signal **121** is described herein. These values are used to compute the measured impedance or dielectric properties over the range of frequencies to correlate with the physical properties of interest for the MUT.

(51) The functionality described herein, or portions thereof, and its various modifications (hereinafter “the functions”) can be implemented, at least in part, via a computer program product, e.g., a computer program tangibly embodied in an information carrier, such as one or more non-transitory machine-readable media, for execution by, or to control the operation of, one or more data processing apparatus, e.g., a programmable processor, a computer, multiple computers, and/or programmable logic components.

(52) A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program can be deployed to be executed on one computer or on multiple computers at one site or distributed across multiple sites and interconnected by a network.

(53) Actions associated with implementing all or part of the functions can be performed by one or more programmable processors executing one or more computer programs to perform the functions of the calibration process. All or part of the functions can be implemented as, special purpose logic circuitry, e.g., an FPGA and/or an ASIC (application-specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. Components of a computer include a processor for executing instructions and one or more memory devices for storing instructions and data.

(54) In various embodiments, components described as being “coupled” to one another can be joined along one or more interfaces. In some embodiments, these interfaces can include junctions between distinct components, and in other cases, these interfaces can include a solidly and/or integrally formed interconnection. That is, in some cases, components that are “coupled” to one another can be simultaneously formed to define a single continuous member. However, in other embodiments, these coupled components can be formed as separate members and be subsequently joined through known processes (e.g., fastening, ultrasonic welding, bonding).

(55) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” may

be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

(56) When an element or layer is referred to as being “on”, “engaged to”, “connected to” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to”, “directly connected to” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(57) Spatially relative terms, such as “inner,” “outer,” “beneath”, “below”, “lower”, “above”, “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(58) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A method comprising: using a measurement system including an electronic circuit to characterize a response signal for detecting physical characteristics of a material under test (MUT), by: transmitting an excitation signal into the MUT and transmitting a reference signal to a set of magnitude and phase (M/P) detectors; receiving the response signal from the MUT based on the excitation signal; comparing a magnitude and phase of the response signal with a corresponding detection range for one of the M/P detectors; comparing a magnitude and phase of the reference signal with a corresponding detection range for each of two of the M/P detectors; and adjusting at least one of the excitation signal or the reference signal based on the comparisons.
2. The method of claim 1, wherein the excitation signal and the reference signal have a common frequency and a distinct magnitude and/or phase.
3. The method of claim 1, wherein adjusting the excitation signal is performed in response to the magnitude and/or phase of the response signal deviating from the corresponding detection range of the one of the M/P detectors.

4. The method of claim 3, wherein adjusting the reference signal is performed in response to the magnitude and/or phase of the reference signal deviating from the corresponding detection range of at least one of the two M/P detectors.
5. The method of claim 3, wherein the excitation signal and the reference signal are both generated by a signal generator with a common control signal.
6. The method of claim 1, further comprising using the measurement system to attenuate the excitation signal prior to comparing the reference signal with the corresponding detection ranges for the two M/P detectors and comparing the response signal with the corresponding detection ranges for the one of the M/P detectors.
7. The method of claim 6, further comprising using the measurement system to down-convert a portion of the excitation signal to a predetermined frequency.
8. The method of claim 7, wherein the predetermined frequency of the down-converted portion of the excitation signal includes a range of approximately 10 kHz to approximately 1 MHz.
9. The method 7, wherein the reference signal remains at a fixed frequency while a portion of the excitation signal is down-converted.
10. The method of claim 9, wherein the fixed frequency matches a frequency of a mixer output.
11. An electronic circuit configured to: apply an excitation signal to a material under test (MUT); receive a response signal from the MUT; determine a magnitude and phase relationship between a response signal produced in presence of the MUT and the reference signal; compute impedance and dielectric properties of the MUT based on the magnitude and phase relationship between the response signal and the reference signal; and apply the computed dielectric properties to a physical model to correlate the measurement to at least one physical property of the MUT.
12. The electronic circuit of claim 11, wherein an amplitude of the excitation signal and an amplitude of the reference signal are independently set to control determination of the magnitude and phase relationship between the response signal and the reference signal.
13. The electronic circuit of claim 11, further configured to: send a first portion of the excitation signal to a magnitude and phase (M/P) detector and send a second portion of the excitation signal to a transmitting electrode for application to the MUT; and attenuate the first portion of the excitation signal to produce an intermediate signal, wherein a frequency of the excitation signal varies across a range of frequencies, and wherein a frequency of the intermediate signal varies with the excitation signal at a lesser amplitude.
14. The electronic circuit of claim 11, wherein computing the impedance and dielectric properties of the MUT is performed using a computational model of physical characteristics of the MUT and a sensing system.
15. The electronic circuit of claim 11, further comprising generating the excitation signal and the reference signal at a specific frequency or over a range of frequencies.
16. A method of using the electronic circuit of claim 11.
17. A sensing system comprising the electronic circuit of claim 11.
18. A method of assembling a system for detecting physical characteristics of a material under test (MUT), the method comprising: connecting an electronic circuit to a user interface and a sensor system; obtaining a physical model of the MUT; using the electronic circuit to compute impedance and dielectric properties of the MUT based on a magnitude and phase relationship between a response signal and a reference signal, wherein the impedance and dielectric properties are computed in a field setting; and applying the computed dielectric properties to a physical model to correlate a measurement to at least one physical property of the MUT, wherein the physical model further defines amplitudes of an excitation signal and the reference signal for correlating the measurement with the at least one physical property of the MUT.
19. The method of claim 18, wherein the sensor system includes an array of electrodes, wherein the array of electrodes includes at least one transmitting electrode and at least one receiving electrode.
20. The method of claim 18, wherein the assembling is performed outside of a laboratory.

